ROBOTICS 4/M

TOPIC 1: INTRODUCTION

1.1 ROBOTICS BACKGROUND

"The fact is, that civilization requires slaves. The Greeks were quite right there. Unless there are slaves to do the ugly, horrible, uninteresting work, culture and contemplation become almost impossible. Human slavery is wrong, insecure and demoralizing. On mechanical slavery of the machine, the future of the world depends."

Oscar Wilde

In 1922 the Czech author Karl Čapek wrote a story called *Rossum's Universal Robots* and introduced the word *robota* to the world [1,2]. This word has been given the meaning *worker* and *slave*. This notion of an automated mechanical device for performing manual labour was first developed by George Devol after significant advances in the field of control [1]. Since then the modern robot has been developed and implemented in a number of different fields e.g. manufacturing, military applications, promotional robots. The majority of the development has been through the manufacturing practices in the automobile industry. This gave rise to a range of *manipulator* or *articulated* (*jointed*) robotic systems [1,2,3]. These manipulator based industrial robots are the most used types. Most robots are derived from these manufacturing systems.

However, the image of the humanoid robot portrayed in the books by *Isaac Asimov* [4] has not become a reality until recently, with the development of humanoid robots by companies like Honda (*ASIMO*), Boston Dynamics (*Atlas*) and Tesla (*Optimus*), among others. This reluctance to develop humanoid systems has been partly due to the fear in humans that robots will replace them [4]. This was particularly the case with the introduction of robots in the manufacturing industry, where reluctance to change to automated systems has always been present.

(a) Honda ASIMO

(b) Boston Dynamics Atlas

(c) Tesla Optimus

Figure 1.1: Humanoid Robots

The introduction of robot manipulators in the manufacturing industries was a result of increased consumer demand. This demand indicated a need for increased productivity in the production line so that the goods can be manufactured more quickly and with improved accuracy. This meant that the production line had to work 24 hours a day and the automated approach was viewed to be the way. Consequently, human manufacturing operatives were replaced by robot manipulators that could meet the production demands imposed by the consumer market. This approach did increase productivity and reduced cost. However, it also reduced the size of the workforce and increased the reluctance to implement and develop robotic systems further.

Asimov's vision of the future foresaw this reluctance of humankind to robots. In order for robots to be viewed as helpers instead of replacement workers, he set for the three basic *Laws* or *Rules for Robotics*. These were:

1. A robot must not harm a human being, nor through inaction allow one to come to harm.
2. A robot must always obey human beings, unless that is in conflict with the first law.
3. A robot must protect itself from harm, unless that is in conflict with the first two laws.

Although these *laws* do go some way to diminishing people's fears by making robots appear subservient to humans, they do not take into account the reluctance of implementing robotic systems for the fear of replacing human workers. To this end, Professor Gerald Norman of Oregon Institute of Technology suggested the introduction of the following fourth rule of robotics [1,2,4]:

4. A robot may take a human being's job. But it must not leave that person jobless.

The robotics community has adopted these rules, particularly the original three, as guidelines for the development of future robotics systems. Naturally, there are certain cases where these laws will be broken e.g. military applications may case for robots to break these laws.

Most industrial robots are designed to replace humans in jobs that are boring due to repetition, unpleasant and hazardous due to chemicals being used, or require an amount of precision that a human is unable to provide. Although most of the systems that you will look at in this course mimic the motion and articulation of humans, it does not need to be a replica of a human or any other natural species e.g. Unmanned Air Vehicles (UAVs) do not replicate the flight of birds. However, there are fields of research where natural species are replicated by robots to provide insight into the species behaviour e.g. robotic fish for freshwater salmon studies [5].

1.2 Classification of Robots

Robots are classified in terms of their capabilities and their interaction with the operator. Fundamentally, these classifications fall into two classifications:

Remotely Operated Robots: The utilisation of these robots is directly controlled by commands from the human operator.

Autonomous Robots: The utilisation of these robots is predetermined by the operator prior to the robot being activated. The robot then executes these defined tasks under its own intelligence.

These classifications have been subdivided by a number of institutions that study robotics. The Japanese Industrial Robot Association (JIRA) defines robots using the following classifications [1]:

Table 1.1: Japanese Industrial Robot Association Classifications

CLASS	DESCRIPTION
1	*Manual-Handling Devices*: A device with multiple degrees of freedom that is controlled by commands from a human operator.
2	*Fixed-Sequence Robot:* A device that performs the successive stages of a task according to a predetermined programme and is difficult to modify.
3	*Variable-Sequence Robot:* A device that performs the successive stages of a task according to a predetermined programme and is easy to modify.
4	*Playback Robot*: A human operator performs a task manually and the robot records the motions involved in the task. The robot repeats these recorded motions.
5	*Numerical Control Robot*: A human operator supplies the robot with a movement programme rather than teaching it manually. The root executes this programme.
6	*Intelligent Robot:* A robot with the means to understand its environment and the ability to successfully complete a task despite changing conditions around it.

In contrast the Robotics Institute of America (RIA) only consider classes 3-6 as robots. The others are regarded as automated devices.

The Association Française de Robotique (AFR) defines robots using the following classifications [1]:

Table 1.2: Association Française de Robotique Classifications

TYPE	DESCRIPTION
A	Handling devices with manual control to remote control.
B	Automatic handling devices with predetermined operation cycles.
C	Programmable, servo controlled robots with continuous or point-to-point trajectories.
D	Same as Type C with the ability to adapt to its environment and any changes that may occur.

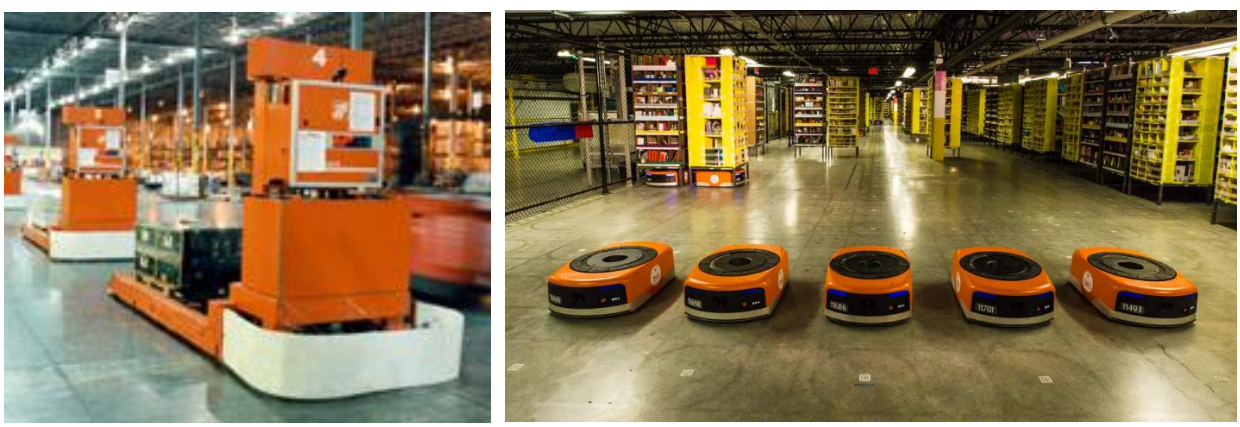
1.3 Types of Robots

There are a number of different types of robots commonly used at present. They can be categorised under the following headings that are related to the robots' applications [2]:

Industrial Robots: These are robots that are intended to perform specific jobs within a manufacturing production line. Industrial robots are intended to perform general purpose unskilled or semiskilled tasks. These robotic systems are usually *fixed base* (i.e. their base is fixed to the floor and the robot is not able to move freely about the factory floor) or *conveyer base* (i.e. the robot is fixed to a conveyer belt system that can move the entire robot along a moveable belt parallel to the line of production). These systems are usually manipulator based (i.e. robot looks like an articulated arm) with specific tools attached to the end of the arm (see Figure 1.2)

Figure 1.2: Industrial Manipulator: Kuka KR 500

As well as manipulators for manufacturing, there are industrial *Autonomous Guided Vehicles* (AGVs) that carry goods and products round a factory (see Figure 1.3).



(a) AGV Products Inc. (b) Amazon's Kiva robots

Figure 1.3: Industrial AGVs

In the remainder of this course we shall concentrate on Industrial Robot Manipulators.

FLYING ROBOTS: The main category of flying robots fall under the heading of *Unmanned Aerial Vehicles* (*UAVs*) or drones (see Figure 1.4).

(a) Eagle Eye Rotorcraft UAV

(b) Fixed wing UAV

Figure 1.4: UAVs

These are any vehicles that can be operated by remote control. The most prominent UAVs are used for reconnaissance, surveying and filming purposes. They can be operated by remote control or totally autonomously (i.e. without human commands after the UAV is activated).

Other types of flying robots are remotely operated weaponry e.g. guided missiles. These are classed as robots within military circles (see Figure 1.5).

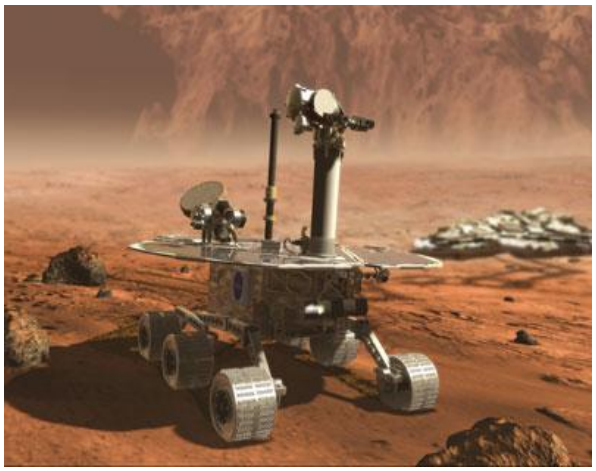
Figure 1.5: Guided Missile: AGM-86 Cruise Missile

SPACE ROBOTS: Space robots are used to provide communications, inspection, and exploration. In terms of communications, remotely operated and semi-autonomous satellites can be used to form a constellation of communication nodes that provide greater coverage. Space robots can be used to inspect space structures and perform duties that may be difficult or hazardous for astronauts to perform. In the case of space structures inspection and maintenance, the robots can perform automated task specified by the operator (see Figure 1.6).

Figure 1.6: Space Robot: SCAMP Camera Platform

In addition to maintenance and inspection, robot probes have been used for many years to explore deep space and other planet surfaces. The most prominent of these are the Voyager and Pathfinder spacecraft that have discovered numerous revelations about the planets of our solar system (see Figure 1.7(a)). The exploration functions of these probes were mostly automated and self-regulated.

(a) Voyager Space Probe



(b) Planetary Exploration Rover.

Figure 1.7: Planetary Exploration Rover

In addition to exploration by space probe, in recent years the surfaces of Mars and the Moon have been explored using *Planetary Exploration Rovers* (PERs) (see Figure 1.7(b)). Current PERs are wheeled robots that are remotely controlled from Earth. However, future developments will involve semi or full autonomous operation.

MARINE ROBOTS: Exploration of the world's oceans has been greatly assisted by surfaces and underwater robots that can journey on top and underneath the water. These underwater robots are equipped with manipulators (i.e. robotic arms) that can be used to perform task underwater. In addition to manipulators and navigation sensors, most Marine Robots have onboard cameras to monitor the path that the robot is taken.

Exploration, monitoring and surveying of the water surface can be carried out by *Autonomous Surface Vessels* (ASVs) or *Unmanned Surface Vehicles* (USVs). As the names suggest, these are uninhabited boats that are able to function under remote control or fully autonomously (see Figure 1.8). They can take the form of any boat (e.g. catamaran, speedboat, sailboat) depending on their operational requirements.

When it comes to underwater robotics there are different types of *Unmanned Underwater Vehicles* (UUVs). The first is the *Remotely Operated Vehicle* (ROV), which is operated by a human via a control panel that is connected to the

(a) C-Target 3 USV

(b) Saildrone sailboat ASV

Figure 1.8: Surface Vessels

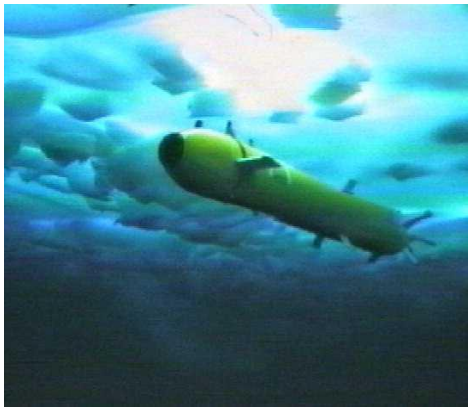
ROV via an umbilical (see Figure 1.9). This basically makes the robot an extension of the operator and a remotely operated tool.

(a) Hyball ROV from Hydrovision Ltd.

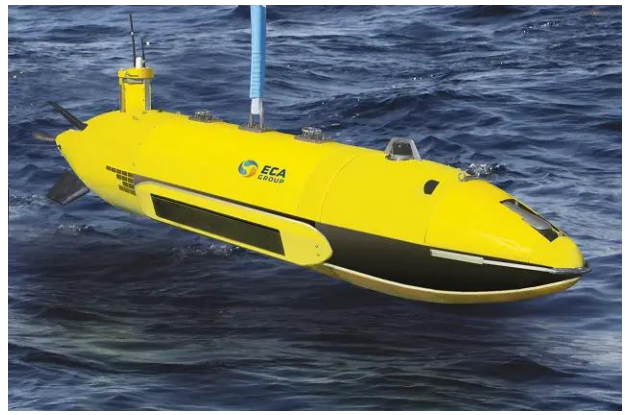
(b) SeaBED

Figure 1.9: ROVs

The second type is the *Autonomous Underwater Vehicle* (AUV), which is preprogrammed by the operator before activated in the water (see Figure 1.10).



(a) Theseus AUV



(b) A27 AUV

Figure 1.10: AUVs

The AUV then executes the programme and performs the task set by the operator (e.g. maintenance of an oilrig, inspection of an underwater structure). Once the AUV has performed this task it returns to a drop off point (either under the water or on the surface) and is retrieved by the operating ship.

The third type are called *Underwater Gliders*, which uses buoyancy and weight for propulsion rather than a propellor (see Figure 1.11).

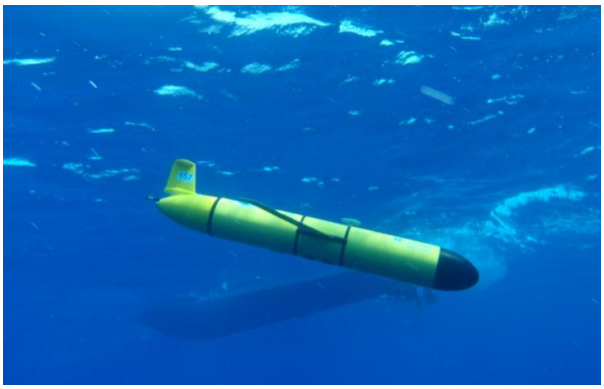


Figure 1.11: Slocum Underwater Glider

MEDICAL AND RESCUE ROBOTS: Medical applications for robots range from keyhole surgery (where the surgeon uses a remotely controlled micro-manipulator with surgical tools at the end) (see Figure 1.12) to robotic systems that can be used to aid, substitute or restore functionality.

Figure 1.12: Keyhole Surgery Robots

These uses of robotic systems have greatly assisted the medical profession. In addition, hospitals use AGVs to perform portering tasks (e.g. moving meals to wards and moving surgical equipment to theatre) (see Figure 1.13).

Figure 1.13: Hospital AGVs

In addition to the use of robots in medical scenarios, an area of advancement is in the field of Rescue Robots. These robots are being developed to explored scenes of accidents (e.g. collapsed buildings), transport survivors and put out fires (see Figure 1.14).

(a) Talon Robot

(b) Medical Recovery Robot

Figure 1.14: Rescue Robots

It is considered that using rescue robots to find possible survivors could greatly improve their chances of recovery and safety.

BIOMIMETIC ROBOTS: As well as functional robots for performing tasks, there is a class of robots that has been designed to mimic biological entities. These Biomimetic Robots are designed to mimic the motion and gait of animals (see Figure 1.15).

(a) Robo-Pike from MIT

(b) Spot the robotic dog

Figure 1.15: Robotic Animals

In addition to animal replicants, there have been a number of human robots developed over recent years (see Figure 1.16). These have managed to capture the gait of humans and thus manoeuvre successfully.

DOMESTIC ROBOTS: In the domestic environment there is already a range of robots. The most popular of these is the range of robotic pets (e.g. AIBO from Sony, see Figure 1.17). These have become very popular over the last few years.

(a) NAO6

(b) NASA's Valkyrie Robot

Figure 1.16: More Humanoid Robots

Figure 1.17: AIBO Robotic Pet from Sony

As well as domestic pets, there are a number of automated household appliances that are regarded as domestic robots. The housekeeping tasks such as carpet cleaning and grass cutting (see Figure 1.18).

(a) Robotic Lawnmower

(b) Roomba Floor Vac Robot

Figure 1.18: Household Robot

These autonomous household robots are able to perform these duties by covering a user-defined area. More intelligent and advanced domestic robots will become available in the near future.

1.4 COMPONENTS OF ROBOTS

Most modern robots consist of the following components that enable them to perform the task that they undertake (see Figure 1.19).

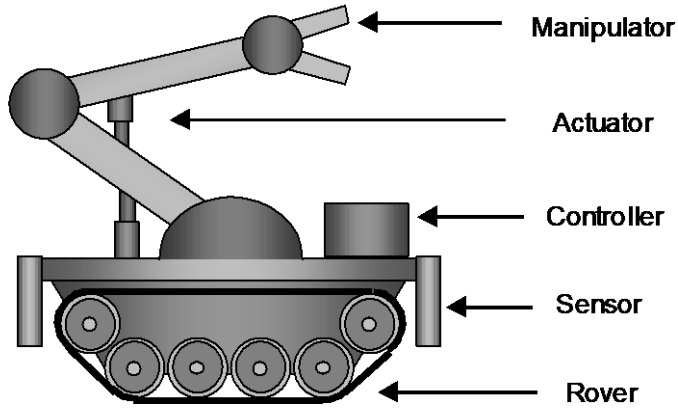


Figure 1.19: Robot Components

MANIPULATOR OR ROVER: These elements constitute the main body of the robot. The manipulator (robot arm) is used to manipulate objects and substances in the robot's external environment. The rover is a mechanism that allows a robot to propel itself through its operating environment (e.g. tracks for travelling over the ground, thrusters for an AUV/ROV). Although these elements constitute the main structure of the robot, without the other elements it cannot be considered a true robot.

ACTUATORS: As you learned in the previous part of the course, Actuators are used to drive the motion of a robot. Commonly used actuators for robots are servomotors, stepper motors, pneumatic and hydraulic cylinders. The actuators are essential components of the robot in order for it to move and manipulate its structure.

SENSORS: The sensors are used by the robot to sense its external environment and its own operation status. The sensor information is used by the robot to determine the correct course of action and motion it should take to perform its task. With intelligent robots the sensor information is also used to adapt to changes in the robot's surroundings.

CONTROLLER: In the context of a robot, the controller is effectively the brain. It is the term used to describe the motion, actuator control based on the execution of the robot's task, the sensor information and the robot's status. These signals are sent to control systems that regulate the operation of the actuators.

The combination of these components gives the robotic system required for the task. So far on this course you have studied actuators and sensors. In the remaining lectures you will study the dynamics of manipulators and the control designs needed to regulate the robot's motion.

In order for any robot to carry out most tasks it must be able to manipulate objects and substances in its external environment. With most robots this is achieved through

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